exercise2

March 29, 2026

1 Exercise 2

[26]:
```
import autograd.numpy as np
```

[35]:
```
csvname = 'multiple_sine_waves.csv'
data = np.loadtxt(csvname,delimiter=',')
x = data[:2,:]
y = data[2:,:]
```

[40]:
```
def feature_transforms(x,w):
    a = w[0] + np.dot(x.T,w[1:])
    return np.sin(a).reshape(1, -1)

def model(x,w):
    # feature transformation
    f = feature_transforms(x,w[0])

    # compute linear combination and return
    a = w[1][0] + np.dot(f.T,w[1][1:])
    return a.T
```

[37]:
```
from autograd import grad

def gradient_descent(g, alpha, max_its, w):
    gradient = grad(g)

    weight_history = [w]
    cost_history = [g(w)]

    for k in range(max_its):
        grad_eval = gradient(w)

        # update each block separately
        w = [
            w[i] - alpha * grad_eval[i]
            for i in range(len(w))
        ]
```

```
        weight_history.append(w)
        cost_history.append(g(w))

    return cost_history, weight_history
```

[38]:
```
def least_squares(w):
        # compute the least squares cost
        cost = np.sum((model(x, w) - y) ** 2)
        return cost / float(y.size)
```

[ ]:
```
# parameter settings
max_its = 2000
alpha = 0.01

# gradient descent call
w0 = [
    np.array([-0.1, 0.1, 0.1]),
    np.array([0.0, 0.1])
]
cost_history, weight_history = gradient_descent(least_squares, alpha, max_its,␣
 ↪w0)
```

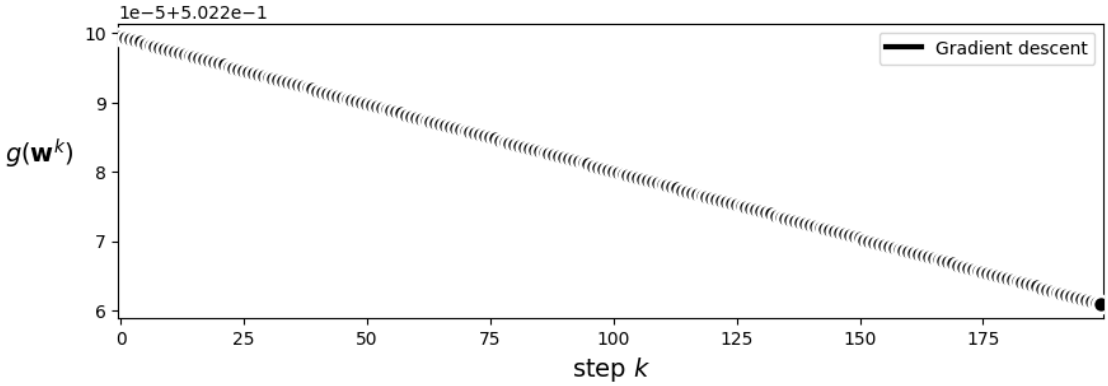
[ ]:
```
# optimal parametters
wopt = weight_history[-1]
wopt
```

[ ]:
```
[array([-0.10527249,  0.05363957,  0.0779151 ]),
 array([-0.05612088, -0.00544886])]
```

The linear parameters are: [-0.10527249, 0.05363957, 0.0779151] The non linear parameters are: [-0.05612088, -0.00544886]

$$\text{yhat} = -0.0561 - 0.00545\,\sin(-0.1053 + 0.0536.\text{x1} + 0.0779.\text{x2})$$

[51]:
```
# cost_history plot
from MLRPlotting import static_plotter
# import custom plotting tools
static_plotter = static_plotter.Visualizer()
static_plotter.plot_cost_histories([cost_history[1800:
 ↪2000]],start=0,points=True,labels=[r'Gradient descent'])
```


1e−5+5.022e−1
10
9
8
7
6
$g(\mathbf{w}^k)$
0
25
50
75
100
125
150
175
step k
Gradient descent